STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-TENSOR-WITH-EXTENSION-BY-ZERO

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1159-1170 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 1172-1199, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_sdga.json (pb=869, stmt 1159-1170, proof 1172-1199).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FR2: sources/stacks-sdga/pinned-original-source.tex, source lines 525–683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E8I: sources/stacks-sdga/pinned-original-source.tex, source lines 1199–1199. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above, let $\mathcal{M}$ be a right graded $\mathcal{A}_U$-module and let $\mathcal{N}$ be a left graded $\mathcal{A}$-module. Then*

$$j_!\mathcal{M} \otimes_{\mathcal{A}} \mathcal{N} = j_!(\mathcal{M} \otimes_{\mathcal{A}_U} \mathcal{N}|_U)$$

as graded $\mathcal{O}$-modules functorially in $\mathcal{M}$ and $\mathcal{N}$.

Proof. Recall that the degree n component of $j_!\mathcal{M} \otimes_{\mathcal{A}} \mathcal{N}$ is the cokernel of the canonical map

$$\bigoplus\nolimits_{r+s+t=n} j_!\mathcal{M}^r \otimes_{\mathcal{O}} \mathcal{A}^s \otimes_{\mathcal{O}} \mathcal{N}^t \longrightarrow \bigoplus\nolimits_{p+q=n} j_!\mathcal{M}^p \otimes_{\mathcal{O}} \mathcal{N}^q$$

See Section 0FR2. By Modules on Sites, Lemma 0E8I this is the same thing as the cokernel of

$$\bigoplus\nolimits_{r+s+t=n} j_!(\mathcal{M}^r \otimes_{\mathcal{O}_U} \mathcal{A}^s|_U \otimes_{\mathcal{O}_U} \mathcal{N}^t|_U) \longrightarrow \bigoplus\nolimits_{p+q=n} j_!(\mathcal{M}^p \otimes_{\mathcal{O}_U} \mathcal{N}^q|_U)$$

and we win. An alternative proof would be to redo the Yoneda argument given in the proof of the lemma cited above. □

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.